

Fractal Dimension of Color Fractal Images

Mihai Ivanovici, *Member, IEEE*, and Noël Richard, *Member, IEEE*

Abstract—Fractal dimension is a very useful metric for the analysis of the images with self-similar content, such as textures. For its computation there exist several approaches, the probabilistic algorithm being accepted as the most elegant approach. However, all the existing methods are defined for 1-D signals or binary images, with extension to grayscale images. Our purpose is to propose a color version of the probabilistic algorithm for the computation of the fractal dimension. To validate this new approach, we also propose an extension of the existing algorithm for the generation of probabilistic fractals, in order to obtain color fractal images. Then we show the results of our experiments and conclude this paper.

Index Terms—Color fractal generation, color images, fractal dimension, fractional Brownian noise, texture.

I. FOREWORD

FRACTAL geometry was introduced by Mandelbrot in 1983 [17] in order to describe self-similar sets called fractals. Fractal dimension is a measure that characterizes the irregularity and the complexity of a fractal, indicating how much space is filled. Lacunarity, a complementary fractal measure defined also by Mandelbrot, indicates how the space is filled. The two properties are usually used as metrics for the discrimination between different structures exhibiting fractal properties, like textures [6], [16].

The fractal-based analysis proved to be of a great interest for the digital image analysis. It is used in an extremely wide-area of applications, such as in finance and stock market, medicine, quality of food analysis, civil engineering, and even art [27]. Its tools, especially the fractal dimension, is used to characterize, classify or to segment images or regions [5], [19], [20], [24].

The fractal-based approaches are perfectly suited for the multiscale analysis of images or any other multidimensional objects, by observing the variation of a measure as a function of the analysis scale. For pure deterministic fractals, the measure is constant in a log-log bidimensional space and, by construction, invariant to scale transformation, rotation or translation of the object. These characteristics justify the choice of this operator in many texture analysis tools.

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M. Ivanovici is with the Transilvania University, Braşov 500036, România (e-mail: Mihai.Ivanovici@gmail.com).

N. Richard is with the University of Poitiers, 86034 Poitiers Cedex, France (e-mail: richard@sic.univ-poitiers.fr).

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Fractal objects are said to be self-similar and independent of scale. The self-similarity is a central concept of the fractal geometry, being closely connected to the notion of dimension. The fractal or similarity dimension is a quantitative measure of the wiggleness of a fractal object [22]. The irregularities of a fractal object determine a fractal dimension comprised in the interval $[E, E + 1]$, where E is the topological dimension of the object. The Hausdorff¹ dimension $H^S(F)$ of a fractal set F is by definition the limit when the diameter δ of the cover set goes to zero

$$H^S(F) = \lim_{\delta \rightarrow 0} H_\delta^S(F) \quad (1)$$

where $H_\delta^S(F)$ is defined as

$$H_\delta^S(F) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^S \right\} \quad (2)$$

U_i is a δ —cover of F , i.e., a countable or finite collection of sets of diameter at most δ that cover the set F .

The Hausdorff dimension is not used in practice, therefore, other equivalent fractal dimensions are defined and used. There are several widely-used approaches for the computation of the fractal dimension: δ -parallel body method (also named covering-blanket approach, Minkowski sausage, or morphological covers) [18], fuzzy [4], [21], pyramidal-decomposition based [2] etc. Most of them are briefly discussed in [13] or in [26]. An attempt to unify several existing approaches into a single one is presented in [15]. Historically, the first practical approach was initiated by Mandelbrot, and clearly expressed by Voss as the probability measure [28], intimately described in [14]. In a parallel way, Allain and Cloitre designed the gliding box-counting algorithm—one of the most used alternative approach, starting from the box-counting algorithm [3]. This approach is further used for the definition of the lacunarity, another widely-used fractal measure. The same gliding box algorithm is used by Plotnick [23] to estimate the fractal dimension and then define and compute the lacunarity of self-similar objects; also mentioned in [10].

Albeit the fact that a plethora of algorithmic approaches exist for the computation of the fractal dimension, only few of them offer the theoretical background that links them to the Hausdorff dimension. From the different expressions directly linked to the theoretical one of the Minkowski–Bouligand [9], the box-counting approach is one of the most popular due to the simplest algorithmic formulation, compared to the Hausdorff dimension. The box-counting fractal dimension is

$$D_{\text{box}} = -\frac{\log N_\delta}{\log \delta} \quad \delta \neq 1 \quad (3)$$

¹also called Hausdorff–Besicovitch dimension.

where $N(\delta)$ is the number of boxes of size δ needed to cover the fractal set.

However, such tools were developed long time ago for grayscale small-size images and they are used *as they are* despite the fact that the acquisition techniques evolved (therefore, the spatial resolution and quantification have changed) and the world of images became colored. There are very few approaches that link the fractal measures to color images and the purpose is only restricted to marginal color analysis and transform a grayscale problem into a false color one [1].

In this paper, we propose to extend the existing probabilistic algorithm by Voss [28] (see Section II) to color images. In order to validate our method, we developed a fractal generator for color images based upon the midpoint displacement algorithm well described in [22] (see Section III). Finally, we present and discuss our experimental results in Section IV and then draw the conclusions.

In addition, we make some remarks on the estimation method of the regression line through the points $(N(\delta), \delta)$ in the log-log space. As expected, the estimation procedure may induce an important variation (see the section of results). All the articles on fractal dimension mention the least square method for the computation of the regression, the correct choice for theoretical fractal objects, but none of them mentions the large variation that may occur (sometimes up to 20%) depending upon the weighting function used in the least-square line-fitting method for natural objects or generated fractals objects.

II. COLOR FRACTAL DIMENSION

The probabilistic algorithm defined by Voss [28] upon the proposal from Mandelbrot [17], chapter 34, considers the image as a set of points S in an Euclidian space of dimension E . The spatial arrangement of the set is characterized by the probability matrix $P(m, L)$, the probability of having m points included into a hyper-cube of size L (also called a box), centered in an arbitrary point of S .

The matrix is normalized so that $\sum_{m=1}^N P(m, L) = 1$, $\forall L$, where N is the number of pixels included in a box of size L . Given the total number of points in the image is M , the number of boxes that contain m points is $(M/m)P(m, L)$. The total number of boxes needed to cover the image is

$$\langle N(L) \rangle = \sum_{m=1}^N \frac{M}{m} P(m, L) = M \sum_{m=1}^N \frac{1}{m} P(m, L). \quad (4)$$

Therefore, $N(L) = \sum_{m=1}^N (1/m) P(m, L) \propto L^{-D}$, where D is the fractal dimension, corresponding to the commonly-used mass dimension. A gray-level image is a 3-D object, in particular a discrete surface $z = f(x, y)$ where z is the luminance in every (x, y) point of the space. There is also an extension of the approach to complex 3-D objects, but its validation is limited to the Cantor dust [8]. However, there is no reference to a development dedicated to color images, despite the fact that the theoretical background for fractal analysis is based upon the Borel set measure in an n -dimensional Euclidian space [9].

A color image is a hyper-surface in a color space, like RGB for instance: $f(x, y) = (r, g, b)$. Therefore, in the case of color



Fig. 1. Three fractal images.

images we deal with a 5-D Euclidian hyper-space and each pixel can be seen as a 5-D vector (x, y, r, g, b) . We have chosen to work in RGB space, not by easiness, but because the RGB space presents a cubic organization coherent with the 2-D spatial organization of the image. In this way, we could meet the constraint of expression in a 5-D space.

The classical algorithm of Voss defines cubes of size L centered in the current pixel $(x, y, z = f(x, y))$ and counts the number of pixels that fall inside a cube characterized by the following corners: $(x - L/2, y - L/2, z - L/2)$ and $(x + L/2, y + L/2, z + L/2)$.

A direct extension of the Voss approach to color images would count the pixels $F = f(x, y, r, g, b)$ for which the Euclidian distance to the center of the hyper-cube, $F_c = f(x_c, y_c, r_c, g_c, b_c)$ would be smaller than L

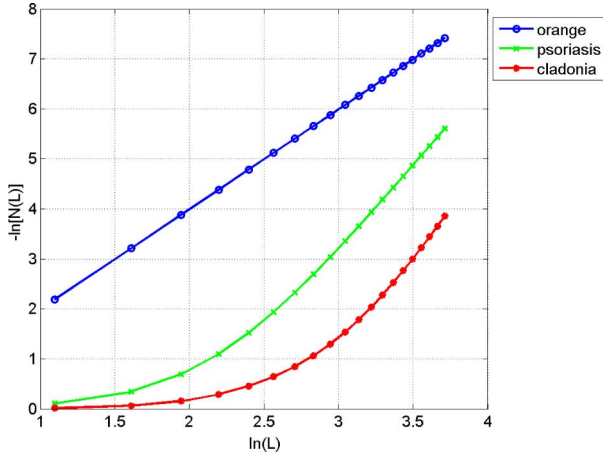
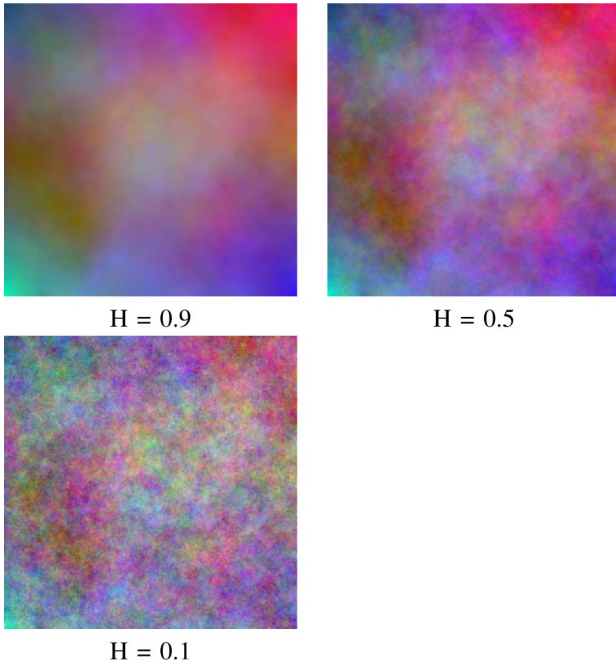
$$|F - F_c| = \sqrt{\sum_{i=1}^5 |f_i - f_{ci}|^2} \leq L. \quad (5)$$

Given that the Euclidian distance in RGB space does not correspond to the perceptual distance between colors, we prefer to use instead the Minkowski infinity norm distance

$$|F - F_c| = \max(|f_i - f_{ci}|) \leq L \quad \forall i = \overline{1, 5}. \quad (6)$$

Practically, for a certain square of size L in the (x, y) plane, we count the number of pixels that fall inside a 3-D RGB cube of size L , centered in the current pixel.

In Fig. 1, we present three images (“orange,” “psoriasis,” and “cladonia”) for which we plot the evolution of $N(L)$ in Fig. 2. One can see that for “orange,” $N(L)$ is a straight line, corresponding to a fractal dimension of 2.0, as expected. For “psoriasis” and “cladonia,” the evolution is no longer linear (see the section of discussion and results for a detailed explanation and the presentation of the solution we propose), which makes the estimation of the regression line slope a more delicate issue.

Fig. 2. $N(L)$ evolution in a log-log space for the images in Fig. 1.Fig. 3. Generated color image fractals for $H = 0.9, 0.5$ and 0.1 , respectively.

III. COLOR FRACTAL GENERATION

Obviously, to write a new fractal estimator is not a great challenge, the more difficult task being to validate the approach on specific objects, for which the theoretical dimension is well known. Unfortunately, a color image does not represent a “standard” five Euclidian form, and a 5-D Brownian fractional noise could not produce a direct stimulus to validate our measure. So to produce a generator not too far from the color texture, we choose to extend the random midpoint displacement method from grayscale [22] to color domain.

The idea of the random midpoint displacement algorithm is to construct the increments in a recursive process, such as the increment is reduced by a factor 2 at each level. To control the texture complexity along the scale, the variance σ_i^2 of the increments must be proportional to $(r^n)^{2H}$ for a Gaussian random variable—the luminance in the case of a grayscale image [22]

$$\sigma_i^2 = \overline{[X(t_1, t_2) - X(s_1, s_2)]^2} \propto \left(\sum_{i=1}^2 (t_i - s_i)^2 \right)^H \quad (7)$$

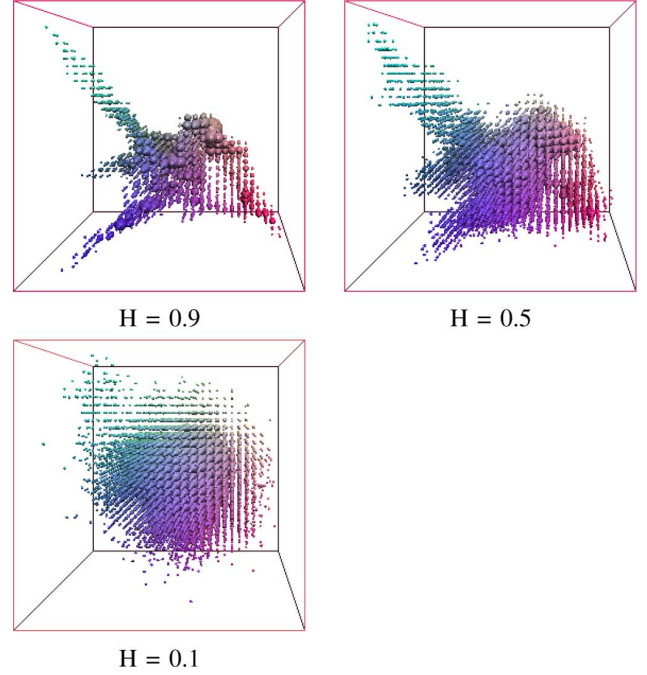


Fig. 4. 3-D color histograms for the generated color fractal images.

where $X(s_i, t_i)$ denotes a random process, a function of two spatial coordinates, s_i and t_i . For a grayscale image, $X(s_i, t_i)$ is the value of the luminance of each pixel of the image, while for a color image is the color of the pixel, thus, a vectorial value. The differences between two random variables are called *increments*. The parameter H —also called the Hurst parameter—characterizes the scaling behavior and it is in the range $0 < H < 1$: a value close to 0 indicates a rough structure, while a value close to 1 indicates a smooth one [22]. The relationship between the fractal dimension D , the dimension of the space E and the parameter H is the following: $D = E + 1 - H$.

In the case of an RGB color image, the increments are differences between two 3-D vectors, in the sense of the Euclidian distance. The variance of these increments is

$$\sigma_i^2 = \overline{\left(\sqrt{\sum_{k=r,g,b} (X_k(t_1, t_2) - X_k(s_1, s_2))^2} \right)^2} \quad (8)$$

$$\sigma_i^2 = \frac{\overline{[X_r(t_1, t_2) - X_r(s_1, s_2)]^2} + \cdots + \overline{[X_g(t_1, t_2) - X_g(s_1, s_2)]^2} + \cdots + \overline{[X_b(t_1, t_2) - X_b(s_1, s_2)]^2}}{3} \quad (9)$$

Given that the increments in the three RGB planes are not correlated, we can distribute the statistical operation to each term

$$\sigma_i^2 = \overline{[X_r(t_1, t_2) - X_r(s_1, s_2)]^2} + \overline{[X_g(t_1, t_2) - X_g(s_1, s_2)]^2} + \overline{[X_b(t_1, t_2) - X_b(s_1, s_2)]^2} \quad (10)$$

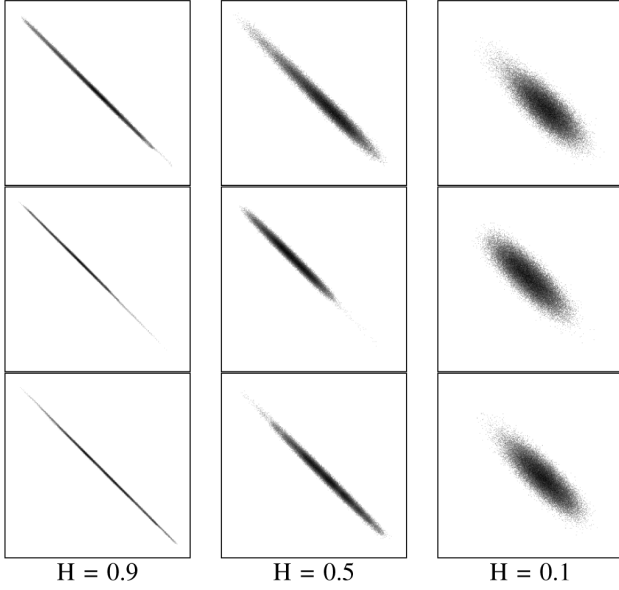


Fig. 5. $M(1,0)$ cooccurrence matrices for the R (top row), G (middle row), and B (bottom row) planes.

Because every variance is proportional to $\left(\sum_{i=1}^2 (t_i - s_i)^2\right)^H$, we can conclude that the sum will be also proportional to the space coordinate differences

$$\sigma_i^2 \propto 3 \cdot \left(\sum_{i=1}^2 (t_i - s_i)^2\right)^H \propto \left(\sum_{i=1}^2 (t_i - s_i)^2\right)^H \quad \square. \quad (11)$$

We, therefore, demonstrated that formula (7) stands for the color fractal images we generated.

We have modified the algorithm presented in [22] in order to work with RGB triplets and we present here only the lines that differ from the original algorithm. In the initialization phase, we generate four RGB triplets for the corners of the fractal image. The `f3` and `f4` functions were modified in order to work with 3-D vectors, as well as the main function which we present in the following:

```
FOR x := 0 TO N STEP D DO
  FOR y := 0 TO N STEP D DO
    X[x][y][r] := X[x][y][r] + delta * Gauss()
    X[x][y][g] := X[x][y][g] + delta * Gauss()
    X[x][y][b] := X[x][y][b] + delta * Gauss()
  END FOR
END FOR.
```

In Fig. 3, three generated color fractal images with different complexity are depicted, for a Hurst factor of 0.9, 0.5, and 0.1, respectively. The size of the images is 256 by 256 pixels, in order to deal with a 5-D Euclidian space that has the same dynamic range on all of its axes.

IV. DISCUSSION AND RESULTS

A. Performance of the Color Fractal Generator

For the color fractal images we computed the 3-D histograms, depicted in Fig. 4. The diameter of each sphere is proportional

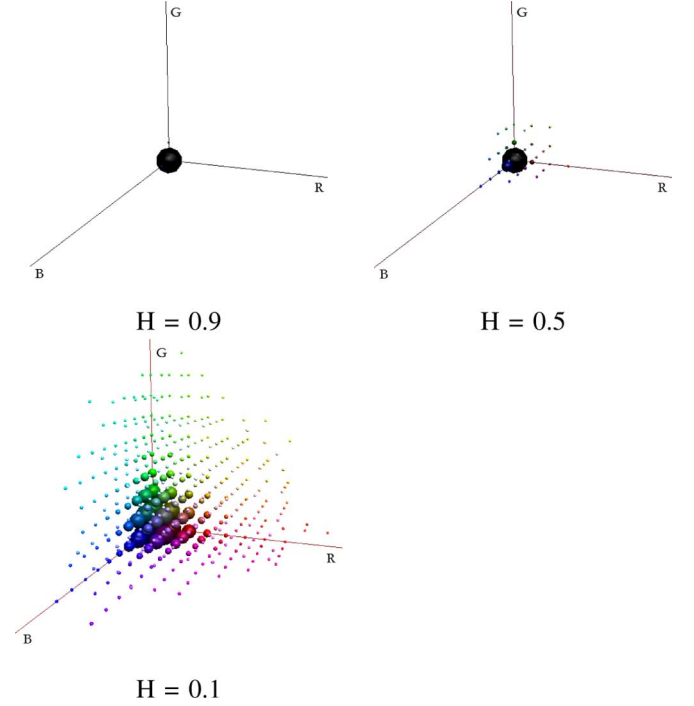


Fig. 6. 3-D histograms of differences between two neighbor color pixels.

to the number of pixels with the same color as the color of the sphere. The histograms are not able to completely characterize the fractal images, but they indicate in an intuitive way how much the RGB is filled. One can see that for $H = 0.9$ the histogram is coarse, being spread in several directions, while for $H = 0.1$ the histogram tends to have the shape of a compact sphere, showing that the RGB space is better filled. The former histogram also shows the limitations of the generator, which is not able of filling the whole RGB space: the larger the occupation of the RGB space, the higher the fractal dimension.

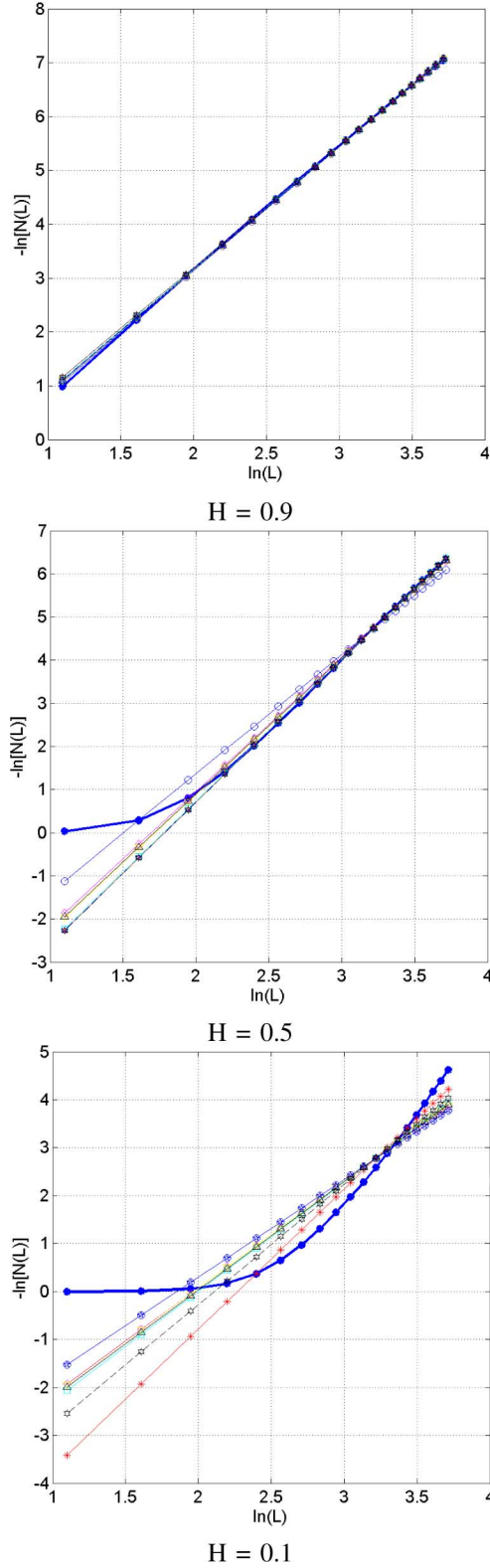
In order to quantify the performance of our color fractal generator, we computed the cooccurrence matrices for a neighborhood distance of 1 pixel for the horizontal direction² [11]. Our purpose is to analyze the repartition of the color increments for the RGB triplets. Theoretically, the cooccurrence matrices for a color image would require a 6-D hyper-cube, and will be impossible to show. But in our case, as the colors increment are uncorrelated between them, we can reduce the computation to the cooccurrence matrix on each color plane of the RGB color space.

The cooccurrence matrices are used to characterize textures by computing the probability $\chi_{c_i, c_j}^{(d, \theta)}$ that a pixel p_1 of color c_i has a neighbor p_2 of color c_j , the position of the neighbor being defined by the distance d and the direction θ

$$\chi_{c_i, c_j}^{(d, \theta)} \triangleq P \left[\frac{p_1 \in \mathbf{A}_{c_i}, p_2 \in \mathbf{A}_{c_j}}{|p_1 - p_2| = d, (p_1, p_2) = \theta} \right]. \quad (12)$$

In Fig. 5, one can see that for the fractal with $H = 0.9$ the cooccurrence matrices exhibit a diagonal shape indicating the small variations between neighbor pixels, while for $H = 0.5$

²In this case, the definition of the cooccurrence is very closed to the correlation except for the limitation imposed by the analysis of the data in a particular direction, which, however, greatly reduces the computation time.

Fig. 7. $N(L)$ as function of L and the regression lines.

and $H = 0.1$ the spread indicates a larger variation. The samples set shape in each plane is clearly a Gaussian and, as expected, the width of this samples set distribution is proportional to the Hurst exponent, which proves the validity of the generator.

TABLE I
FRACTAL DIMENSION FOR THE THREE COLOR FRACTAL IMAGES, DEPENDING UPON THE FLAVOR OF THE LEAST-SQUARE METHOD

method	H = 0.9	H = 0.5	H = 0.1
ols	2.301	2.758	2.032
andrews	2.253	3.303	2.923
bisquare	2.253	3.303	2.923
cauchy	2.275	3.284	2.288
fair	2.287	3.119	2.222
huber	2.286	3.163	2.230
logistic	2.284	3.154	2.255
talwar	2.279	3.300	2.032
welsch	2.254	3.304	2.517
mean	2.275	3.187	2.380
variance	0.0003	0.0316	0.1153

TABLE II
PARTS OF THE PROBABILITIES TABLE FOR GENERATED IMAGE OF $H = 0.1$

	3	5	7	9	11
1	1.00	0.98	0.89	0.71	0.48
2	0.00	0.02	0.10	0.23	0.28
3	0.00	0.00	0.01	0.05	0.14
4	0.00	0.00	0.00	0.01	0.06
5	0.00	0.00	0.00	0.00	0.02
6	0.00	0.00	0.00	0.00	0.00

To complete the previous remarks, we analyzed directly the vectorial (r, g, b) increments and in Fig. 6 we plot the 3-D histograms of the differences between a pixel and its right neighbor. Clearly this is not a marginal analysis (plane by plane) and those three increment histograms show the dependence of the color increments to the Hurst coefficient, and the Gaussian shape of the represented distribution.³

B. Calibration of the Color Fractal Dimension Estimator

In order to validate our approach, we used synthetic fractals generated by using the method described in Section III, for which we demonstrated the increasing complexity in the RGB color space. In Fig. 7, we represented $N(L)$ as a function of L in a log-log space (blue thick starred line). On the same figures we plot the regression lines obtained from different methods to estimate the fractal dimension of the three fractals, for $H = 0.9$, 0.5 and 0.1 , respectively.

There exist several approaches to the linear regression computation, all based upon a iteratively reweighted least-square algorithm. Since Mandelbrot, in all the cited articles, only the “ordinary” least squares method (*ols*) is mentioned for the estimation of the slope of the log-log curves. In this work, we used nine methods for linear regression on noisy datasets, all available in the Matlab Statistics Toolbox. The difference between the nine methods consists mainly in the weighting function used for the computation of the residuals at each iteration, and implicitly to give lower importance to the points that do not fit “well.”

At a first glance, the generated complexity in images is well captured by our color fractal measure, but the estimated fractal dimension is not well correlated to the generated one for all linear regression approaches (see Table I). In particular, when the complexity of the fractal grows up, the obtained $N(L)$ curve

³As an alternative, the two ways of statistical analysis presented so far could constitute another approach to the estimation of D or H , but this does not make the subject of the current article.

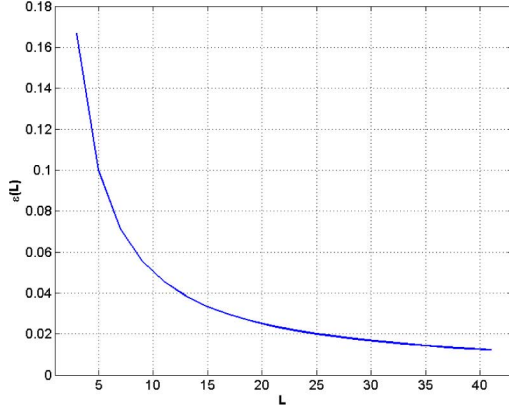


Fig. 8. Maximum quantization error as a function of the box size.

bends, diverting from a straight line. If we were to jump the conclusions, we would say that the analyzed object is not a fractal! Although the estimated fractal dimension does not agree with the increased image complexity, one can notice that the mismatch is due to the poorly calculated slope of the regression line. In addition, for large scales, the $N(L)$ curves indicate the correct ranking.

A thorough introspective indicates that for the generated fractals of middle and high complexity, the measure is altered for the small scales: the emerging “perturbations” imply the fact that the fractal dimension D' restricted to those scales tends to be close to zero, which is clearly false

$$\text{if } \left(-\frac{\log N(L)}{\log L} \right) = 0 \Rightarrow \log N(L) = 0$$

$$\text{and so } N(L) = 1 \quad (13)$$

$$\text{as } N(L) \propto L^{-D'} \Rightarrow L^{-D'} = 1$$

$$\text{then } \Rightarrow D' = 0! \quad (14)$$

As $N(L)$ is defined as

$$N(L) = \sum_{m=1}^N \frac{1}{m} \cdot P(m, L) \quad (15)$$

there are many possibilities for m and $P(m, L)$ so that $N(L) = 1$, but investigating the values in the probability matrix $P(m, L)$ (Table II), the case we are dealing with is particular i.e., $P(m, L) = 1$ with $m = 1$ and $P(m, L) = 0 \forall m \geq 2$.

What is the explanation for this apparently especial situation? If $P(1, L)$ equals one or is very close to one, than this is an indication that the hyper-cube of size L is inappropriate for measuring the fractal dimension, in other words is not able to capture the variations in the fractal image. Those points will lead to a small fractal dimension and will also introduce a larger variation in the estimation of the slope of the regression line. So the different linear regression approaches could not solve themselves this problem, which is more complex than a color noise addition to the measure.

There are two ways to solve this problem: 1) to modify the algorithm; unconsciously, it is the reason for different variants in the box-counting approaches (like differential box-counting [25] and shifting box-counting [6]); and 2) to weight the sample values according to their expected accuracy.

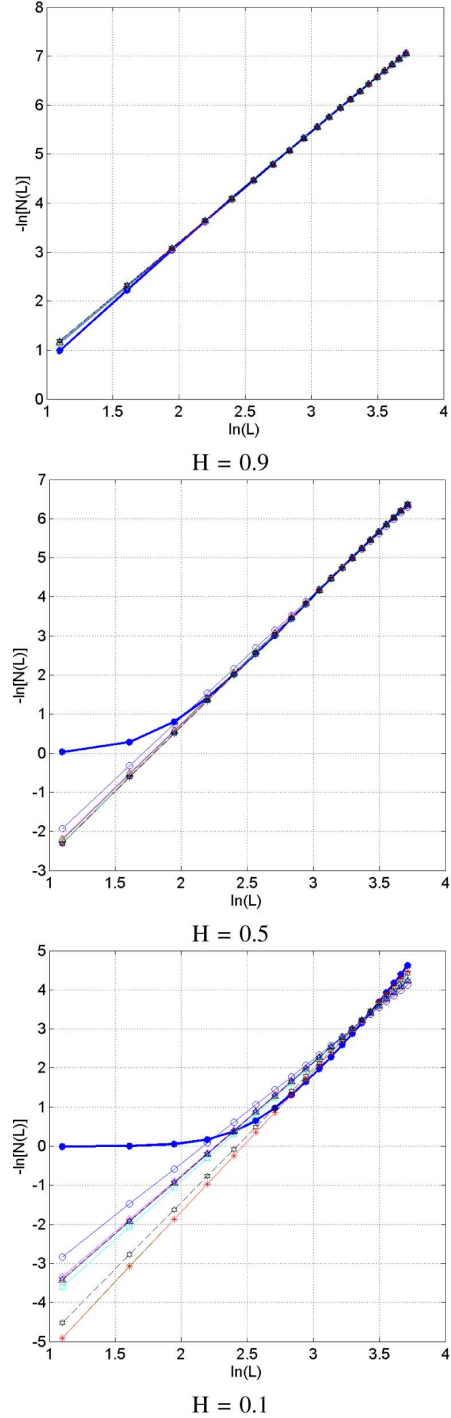


Fig. 9. $N(L)$ as function of L and the regression lines for w_1 .

As expressed before, for fractal image of high complexity the dimension measures are less affected by the quantization error for medium and large box sizes than for the small ones. Therefore, it seems natural that the points corresponding to the medium and large boxes to be taken into account more than the ones corresponding to the small boxes. In case of a grayscale image the maximum quantification error for a pixel is one given that the current value may have a correspondence in the continuous domain that varies within the interval ± 0.5 . For a cube of size $L = 3$, the maximum number of pixels that may fall into it is

TABLE III
FRACTAL DIMENSION FOR ALL THE NINE METHODS AND FOR THE FIRST
PROPOSED WEIGHTING FUNCTION

$w(L)$	$w_1(L) = 1 - \varepsilon(L)$		
method	H = 0.9	H = 0.5	H = 0.1
ols	2.271	3.145	2.661
andrews	2.245	3.324	3.591
bisquare	2.245	3.324	3.586
cauchy	2.252	3.309	3.004
fair	2.262	3.268	2.891
huber	2.258	3.280	2.926
logistic	2.259	3.280	2.927
talwar	2.243	3.320	2.918
welsch	2.245	3.321	3.417
mean	2.253	3.286	3.102
variance	0.000096	0.0033	0.1147

TABLE IV
FRACTAL DIMENSION FOR ALL THE NINE METHODS AND FOR THE SECOND
PROPOSED WEIGHTING FUNCTION

$w(L)$	$w_2(L) = \frac{1}{\varepsilon(L)}$		
method	H = 0.9	H = 0.5	H = 0.1
ols	2.273	3.059	2.713
andrews	2.238	3.312	3.837
bisquare	2.238	3.311	3.837
cauchy	2.243	3.305	3.583
fair	2.252	3.269	3.186
huber	2.247	3.275	3.264
logistic	2.248	3.277	3.267
talwar	2.240	3.264	3.643
welsch	2.239	3.310	3.828
mean	2.246	3.264	3.462
variance	0.00012	0.0064	0.1472

$L^2 = 9$, therefore, the absolute maximum error is $9 \times 0.5 = 4.5$. By extrapolation and dividing the maximum error by the volume of the cube, the maximum quantization error will be given by

$$\varepsilon(L) = \frac{0.5L^2}{L^3} = \frac{0.5}{L}.$$

For a color image, the maximum number of pixels that fall inside a hyper-cube remains the same, except for the fact that the error triples with the plane count. The dependency of the quantization error on the size of the boxes is depicted in Fig. 8.

We, therefore, propose to use the following weighting functions:

$$w_1(L) = 1 - \varepsilon(L) \quad w_2(L) = \frac{1}{\varepsilon(L)} \quad w_3(L) = \frac{1}{\varepsilon^2(L)}.$$

In Figs. 9–11, one can see the impact of these three weighting functions in the estimation of the regression line (see also Tables III–V). As expected, the estimated slope is more linked to the middle and high scales of the fractal measure.

The fractal dimension obtained after weighting the measured data exhibits an increased range of values (up to 4.066 for w_3), as well as a smaller variance for the results of the nine estimation methods. Now for the modified linear regression method, the maximum estimated dimension reaches a value greater than 3.0, and for three estimation methods (andrews, bisquare and welsch) a value larger than 4.0. For this reason, we propose to use one of these three methods to discriminate the color textures by using the proposed color fractal dimension. The third

TABLE V
FRACTAL DIMENSION FOR ALL THE NINE METHODS AND FOR THE THIRD
PROPOSED WEIGHTING FUNCTION

$w(L)$	$w_3(L) = \frac{1}{\varepsilon^2(L)}$		
method	H = 0.9	H = 0.5	H = 0.1
ols	2.251	3.213	3.279
andrews	2.233	3.249	4.066
bisquare	2.233	3.251	4.066
cauchy	2.234	3.279	3.895
fair	2.239	3.274	3.712
huber	2.237	3.272	3.773
logistic	2.238	3.274	3.770
talwar	2.233	3.241	3.974
welsch	2.233	3.276	4.023
mean	2.237	3.258	3.840
variance	0.000035	0.00048	0.0621

TABLE VI
MARGINAL FRACTAL ANALYSIS FOR THE IMAGES IN FIG. 1

image	gray-scale	red	green	blue
“psoriasis”	2.707	2.713	2.700	2.706
“cladonia”	2.527	2.536	2.551	2.479

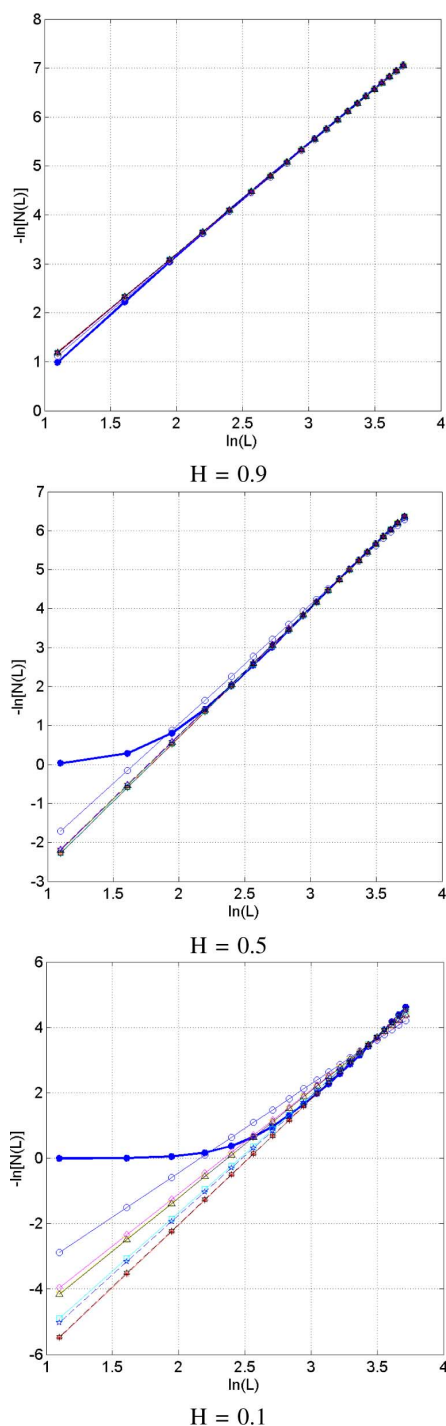
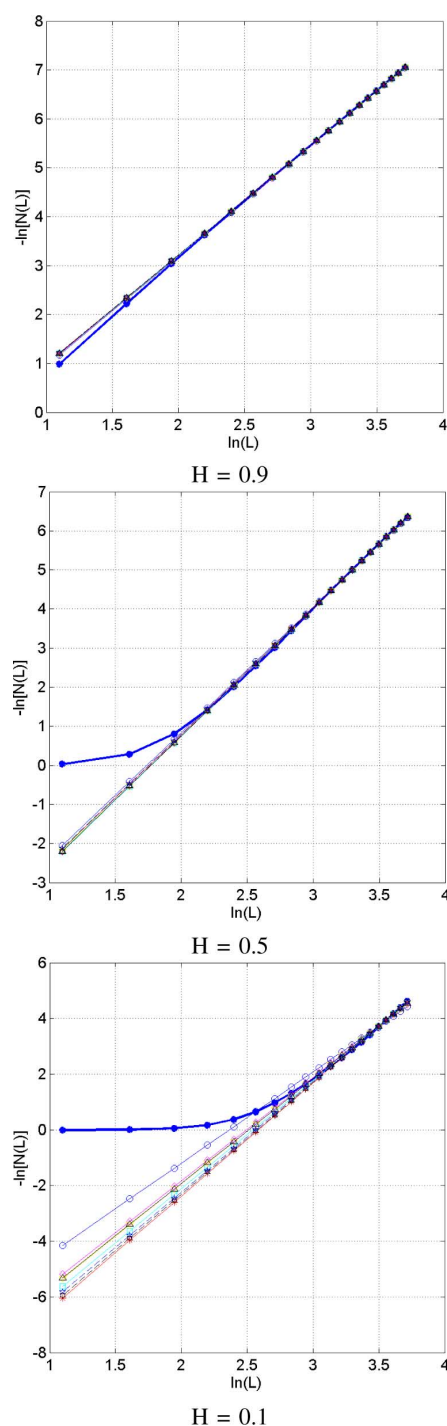
weighting function is more effective than the former two in rejecting the points corresponding to the small scales and consequently leads to steadier results.

For the images presented in Fig. 1 we obtain the following color fractal dimensions: 2.00, 3.39, and 3.71 for “orange,” “psoriasis” and “cladonia,” respectively, by using the third weighting function and the andrews and the bisquare regression. For the same images, we present in Table VI the results of a marginal analysis, i.e., the fractal dimension computed for the grayscale images, and for each of the three RGB planes. From an RGB color point of view, definitely the “cladonia” image is more complex than the “psoriasis” image, but this fact cannot be captured by a marginal analysis, which surprisingly enough indicates the contrary.

V. CONCLUSION

In this paper, we proposed a color extension of the classical probabilistic algorithm, widely used to compute the fractal dimension of images, as well as a method for generating color fractal images. Theoretically, for a color image, i.e., a 5-D fractal, the fractal dimension is comprised between two (uniform color) and five (the highest possible complexity). The fractal dimension of 4.0 we measure indicates the fact that our algorithm it is able to capture the complexity of a color image. A fractal dimension larger than four may imply that for a given spatial position more than one color pixel should coexist.

We demonstrated that our color fractal images are coherent with the existing methods for the generation of probabilistic fractals and showed that the fractal dimension we computed for the color fractal images reflects the complexity in the RGB space, despite the fact that apparently the images may seem visually more complex. Obviously, the probabilistic color midpoint displacement method does not generate the expected complexity, but nevertheless the cooccurrences matrix analysis show the correct ranking of generated fractal images and so the validity of our fractal estimator.

Fig. 10. $N(L)$ as function of L and the regression lines for w_2 .Fig. 11. $N(L)$ as function of L and the regression lines for w_3 .

In addition, we drew the attention on several issues related to the computation of the fractal dimension, such as the different response of the different weighting functions for the line fitting method and the fact that not all hyper-cube sizes are suited for the analysis of a particular image. We have proposed three weighting functions for the importance of the points in the log-log space and proved their usefulness to enhance the range of fractal dimension estimation. This modification on a simple least square approach brings an answer to Huang in the estimation of the fractal dimension [12]. Nevertheless, the weighting

function is strictly dependant upon the choice of the fractal measure and the proposed weighting functions in particular are not suited for other methods like Covering-Blanket or frequential approaches.

To conclude, our purpose was to develop a texture analysis for color images method based upon fractal analysis. The perceived complexity of a color texture can be lower than the one revealed by a marginal (plane by plane) analysis. Because the great majority of images are uncalibrated and stored without information about the illuminant or the acquisition conditions, the best

color space for analysis is the RGB space or another derivate by linear space transformation [7]. In addition, we made the choice of image representation in the RGB space due to the coherence between the generation method and the analysis approach. Nevertheless, this choice is not a constraint and if necessary, the fractal measure can be expressed by using other color spaces, the only issue being then the distance expression between the color components and the spatial coordinates.

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Mihai Ivanovici (M'09) received the B.Sc. degree from the Transilvania University, Braşov, România, in 2001, and the M.Sc. and Ph.D. degrees from the Politehnica University, Bucureşti, România, in 2002 and 2006, respectively.

Currently, he is a Lecturer at the Faculty of Electrical Engineering and Computer Science, Transilvania University, Braşov, România. He is an Associate Member of the Image Processing and Analysis Laboratory from Politehnica University, Bucureşti, România. The research for his Ph.D.

thesis was carried out at CERN, Geneva, Switzerland, from 2002 to 2005, where he was a Project Associate. In 2007 and 2008, he was a Postdoctoral Researcher at the Signals, Images, and Communications (SIC) Laboratory, University of Poitiers, Poitiers, France. His research interests include digital signal and image processing and analysis, and applications in biology and medicine.



Noël Richard (M'08) received the M.Sc. and Ph.D. degrees from the University of Poitiers, Poitiers, France, in 1988 and 1993, respectively.

Since 1992, he has been a Researcher at XLIM-SIC (Signals, Images and Communications) UMR CNRS 6172 Laboratory. He manages several regional projects on the cultural heritage through digital image processing and knowledge-based systems. His research interests include image analysis by means of fractal geometry, color and texture features. Since 2000, his work focuses on the color

perception of images and texture in human vision and the link between images and semantics through ontology.